Date: 06/05/2026

Duration: 2 hours

MONTHLY CHALLENGE

III YEAR – SEMESTER 6

SET C

Question 1:

1. Problem Title & Link

Title: Partition Array into K Subsets with Minimum Maximum Sum

2. Problem Statement

You are given:

- An integer array nums
- An integer k

You must divide the array into **k non-empty subsets** such that:

- Every element is used exactly once
- Subsets can have **different sizes**

Goal

- Minimize the **maximum subset sum** among all subsets

- Return that **minimum possible maximum sum**

3. Examples (Input → Output)

Example 1

Input:

nums = [1,2,3,4]

k = 2

Output:

5

Explanation:

[1,4] and [2,3] → max = 5 (optimal)

4. Constraints

- 1 <= len(nums) <= 16
- 1 <= k <= len(nums)
- 1 <= nums[i] <= 10^4

Question 2:

1. Problem Title & Link

Title: Minimum Cost to Convert String with Transform Rules

2. Problem Statement

You are given:

- A string source
- A string target (same length as source)
- A list of transformations:

from[i] → to[i] with cost[i]

Transformation Rule

- You can convert any character in source using given rules
- Each transformation has a **cost**
- You can apply transformations **multiple times** (chain allowed)

Goal

- Convert source → target with **minimum total cost**

- If impossible: return -1

3. Examples

Example 1

source = "abc"

target = "bcd"

from = ["a","b","c"]

to   = ["b","c","d"]

cost = [1,1,1]

Output: 3

4. Constraints

- 1 <= len(source) <= 10^5
- Characters are lowercase English letters
- 1 <= len(from) <= 100
- Costs are positive